Sparksheet - Transforming Spreadsheets into Spark Data Frames

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Universidad del Valle de Guatemala

SPARK SUMMIT EUROPE 2016

About

- Researcher at Universidad del Valle de Guatemala.

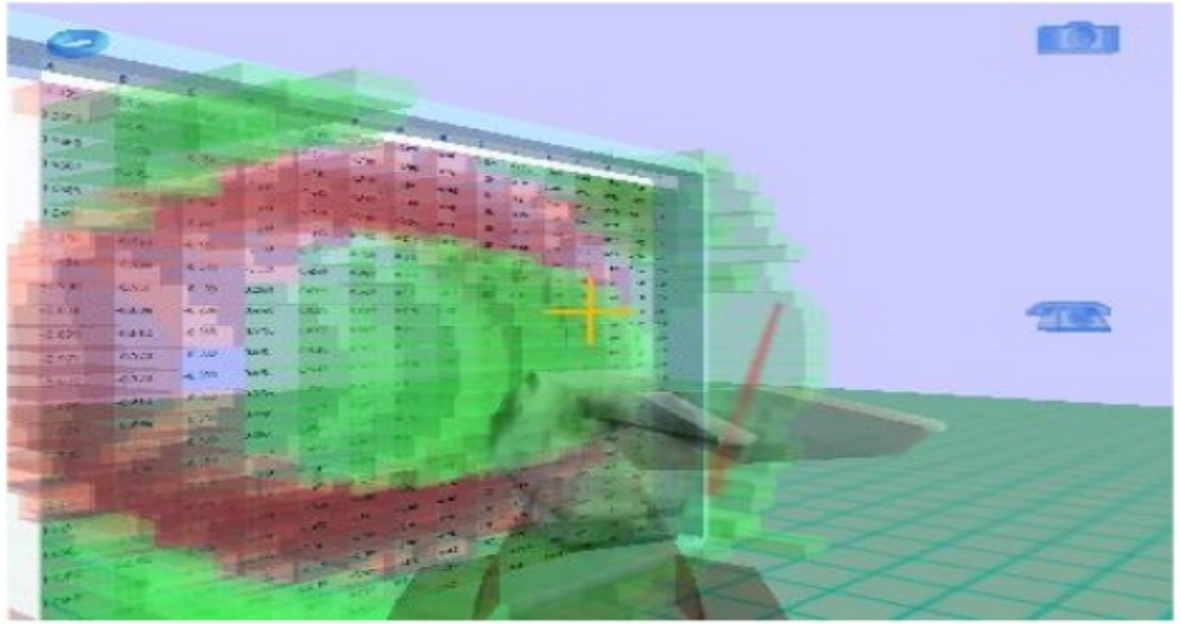
- Research Interests:
  - Program Transformation,
  - Programming Education Research,
  - Online Learning to Rank.

Prototyping …

http://bit.ly/2p5GmvY

Prototyping Spark programs with …

http://bit.ly/2o5Gmy Y



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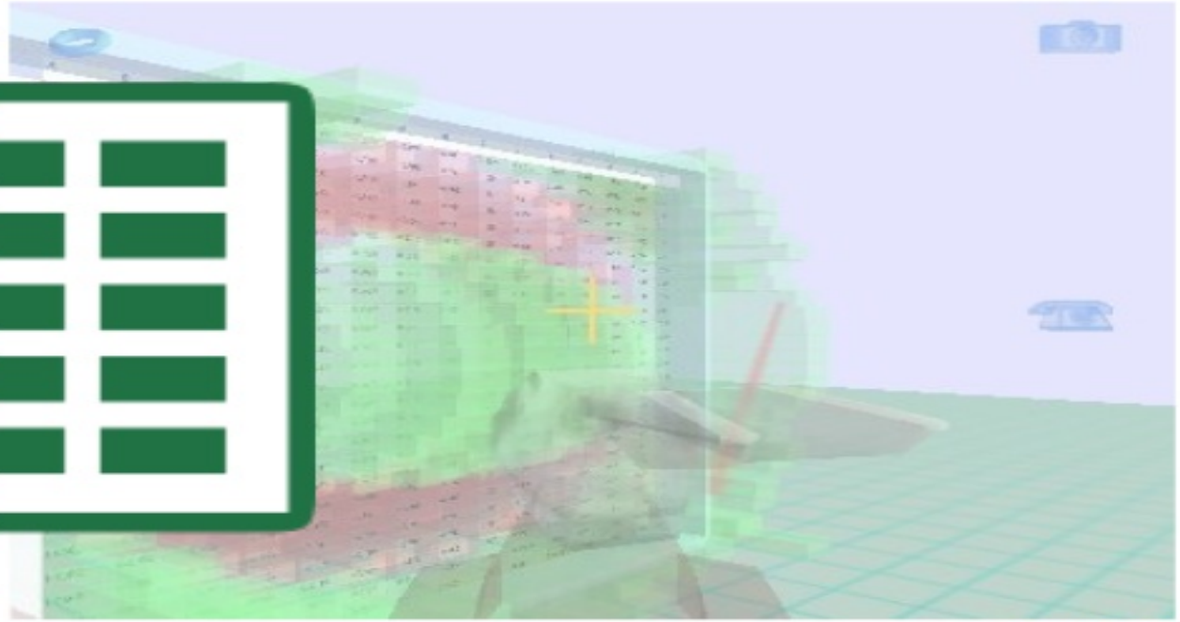
Prototyping Spark programs with Excel



http://bit.ly/2o5GmvY

http://bit.ly/2o0TZA8

http://bit.ly/2odYfMs

Agenda

- Problem Statement and Motivation
  - Architecture
- Program Transformation
  - Pipeline
  - Code-to-Code Transformation
- Parsing Excel Formulas
  - Grammar
  - Parse Tree
  - XLParser
- Excel as a DSL
  - Generating Code
- Demo
- Q&A

Disclaimer(s)

- Ongoing research …
- We will focus on how to create a Program Transformation Pipeline.

Problem Statement

Spark programs can be prototyped in Excel but manually translating Excel formulas to Spark programs is tedious and error-prone.

Motivation

- "***Straight path***" between column-oriented Excel "programs" and Spark programs that make use of the DataFrame API.
- But, manually translating Excel formulas to Spark is tedious and error-prone.
- **What if**: Excel compiler?

Problem Statement

Given that column-oriented Excel applications can be manually translated to Spark programs, …

… find a way to automate translation of Excel formulas so that data pipelines can be prototyped in Excel …

… and Scala/Python code generated to run in Spark.

Motivation

Automatically translate
Excel Formulas to …

Spark.

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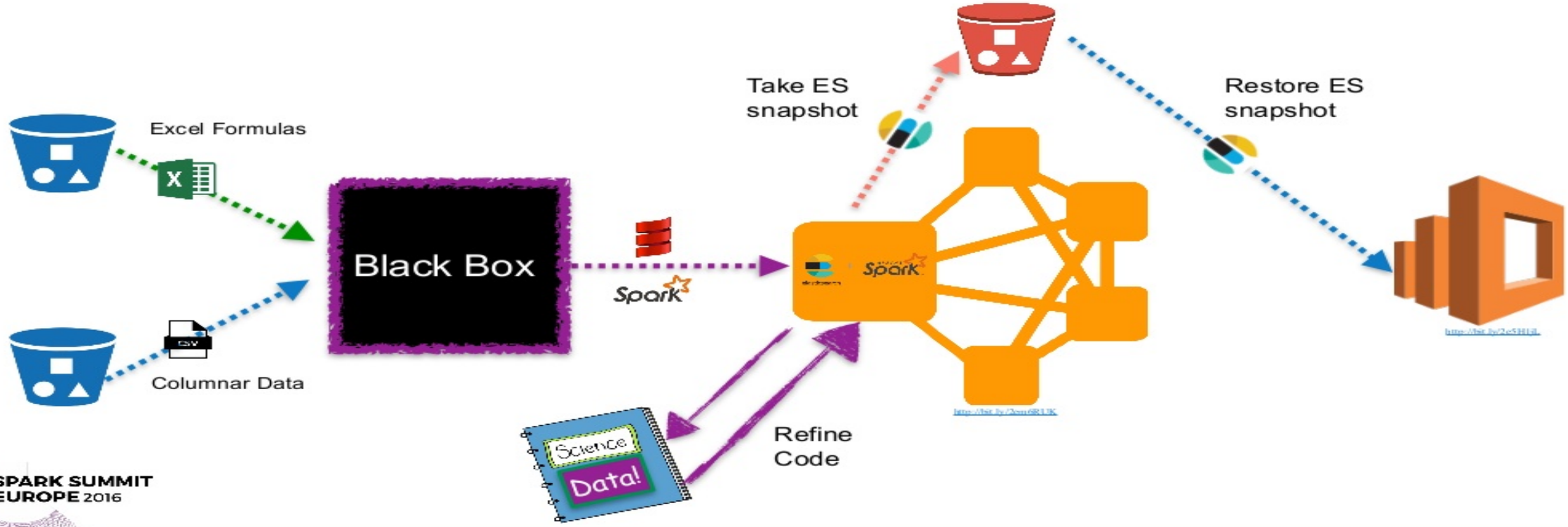
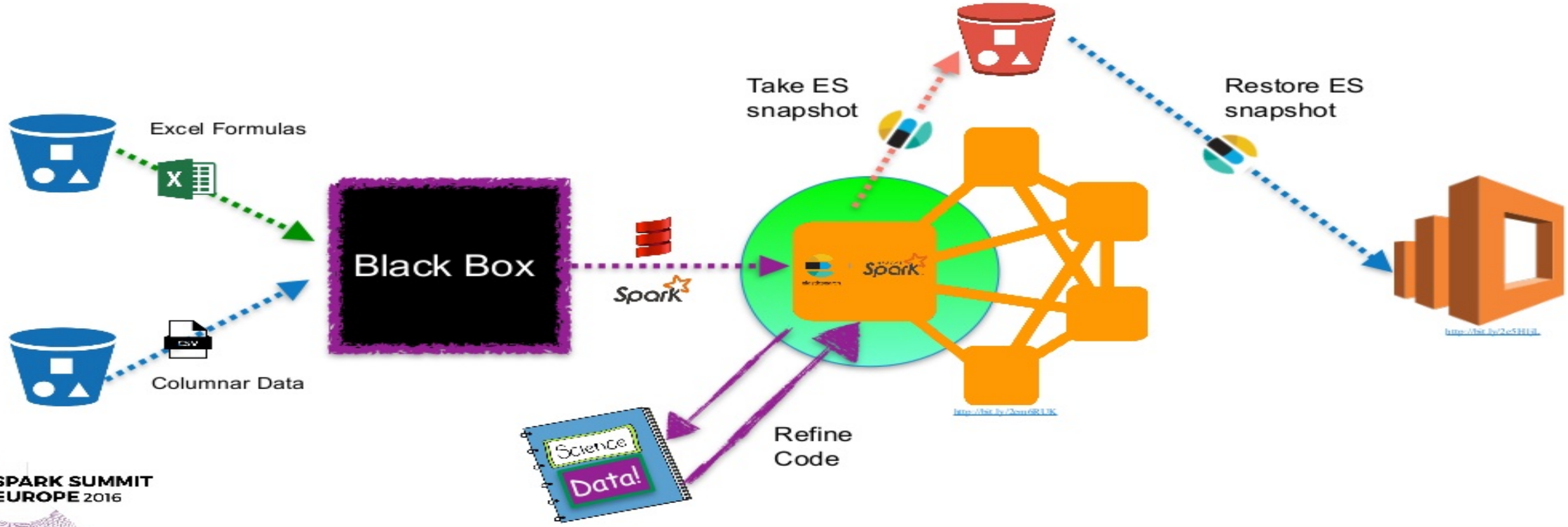
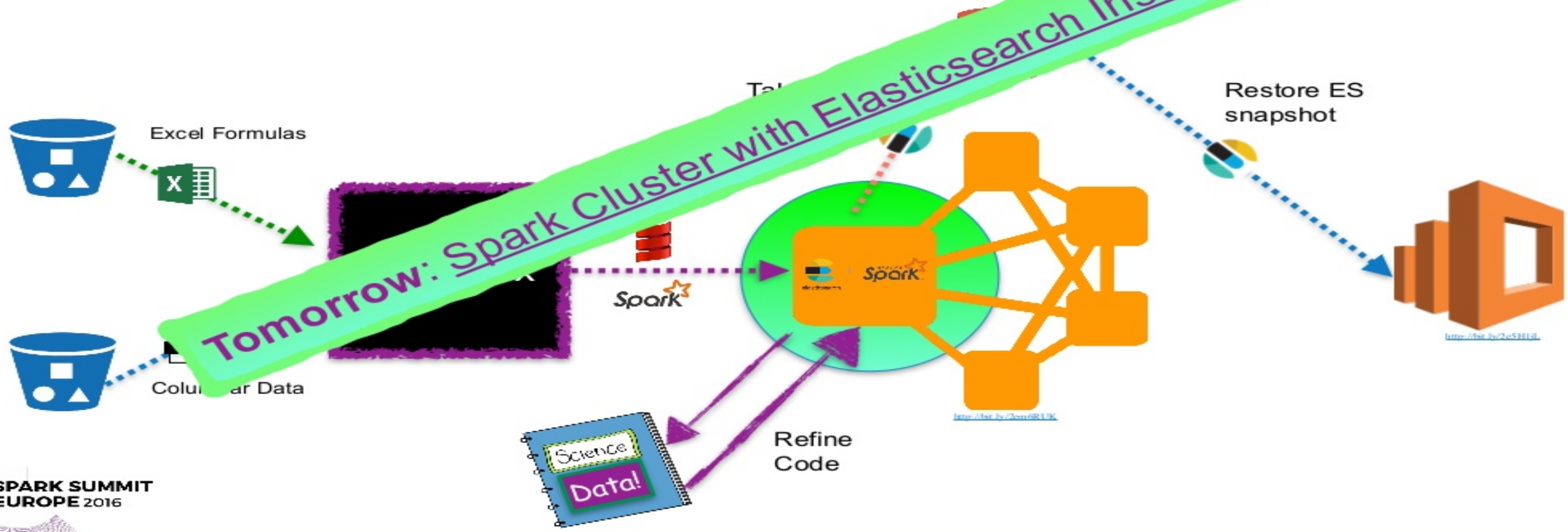
Architecture



Architecture



Architecture



Tomorrow: Spark Cluster with Elasticsearch Inside

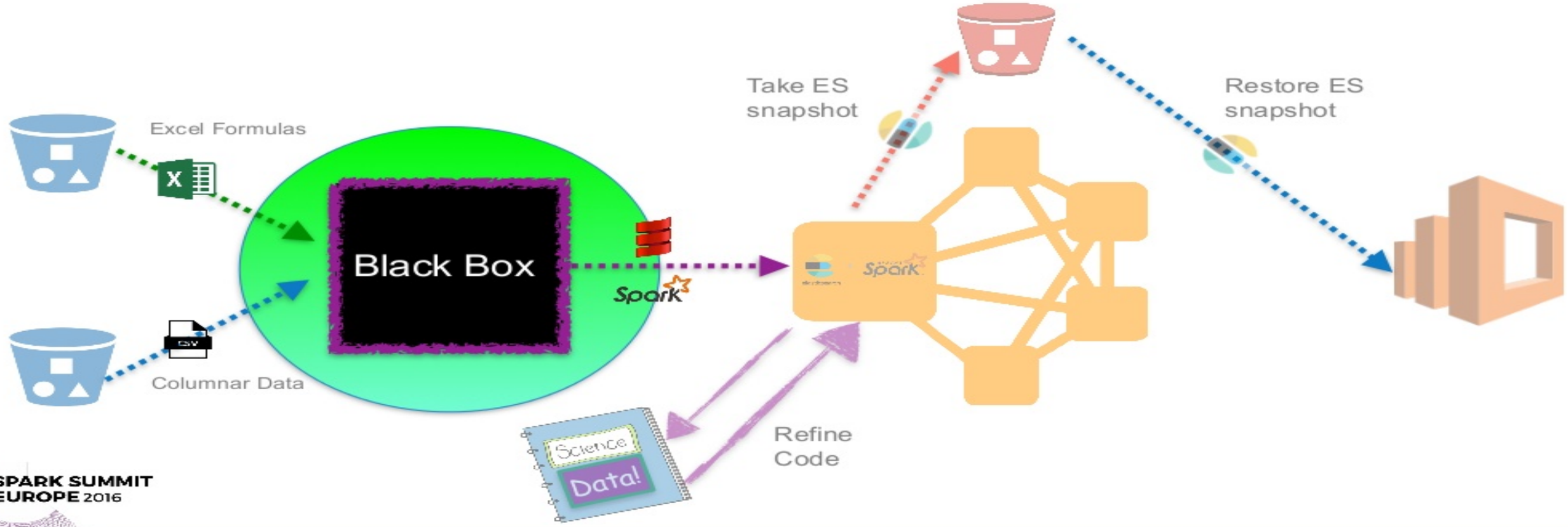
Program Transformation

Program Transformation

"A **program transformation** is any operation that takes a computer program and generates another program."

https://en.wikipedia.org/wiki/Program_transformation

Architecture



Program Transformation Pipeline

http://bit.ly/2eOTZA8

http://bit.ly/2e6alh4

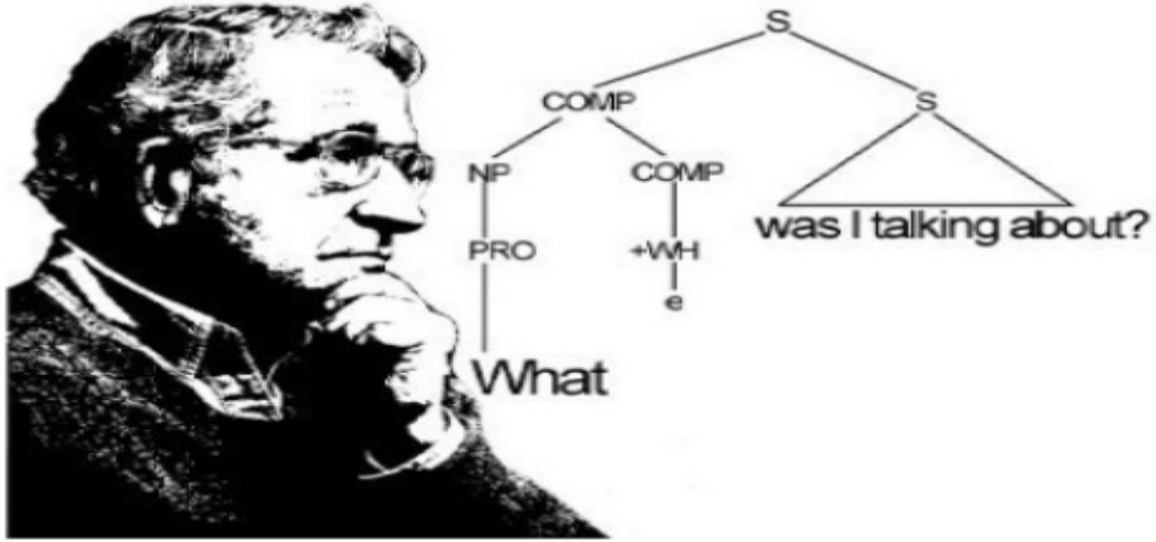
http://bit.ly/2dl0cEq

Code-to-Code Transformation

"The input to the code generator typically consists of a **parse tree** or an **abstract syntax tree**."

https://en.wikipedia.org/wiki/Code_generation_(compiler)

We need a Grammar!



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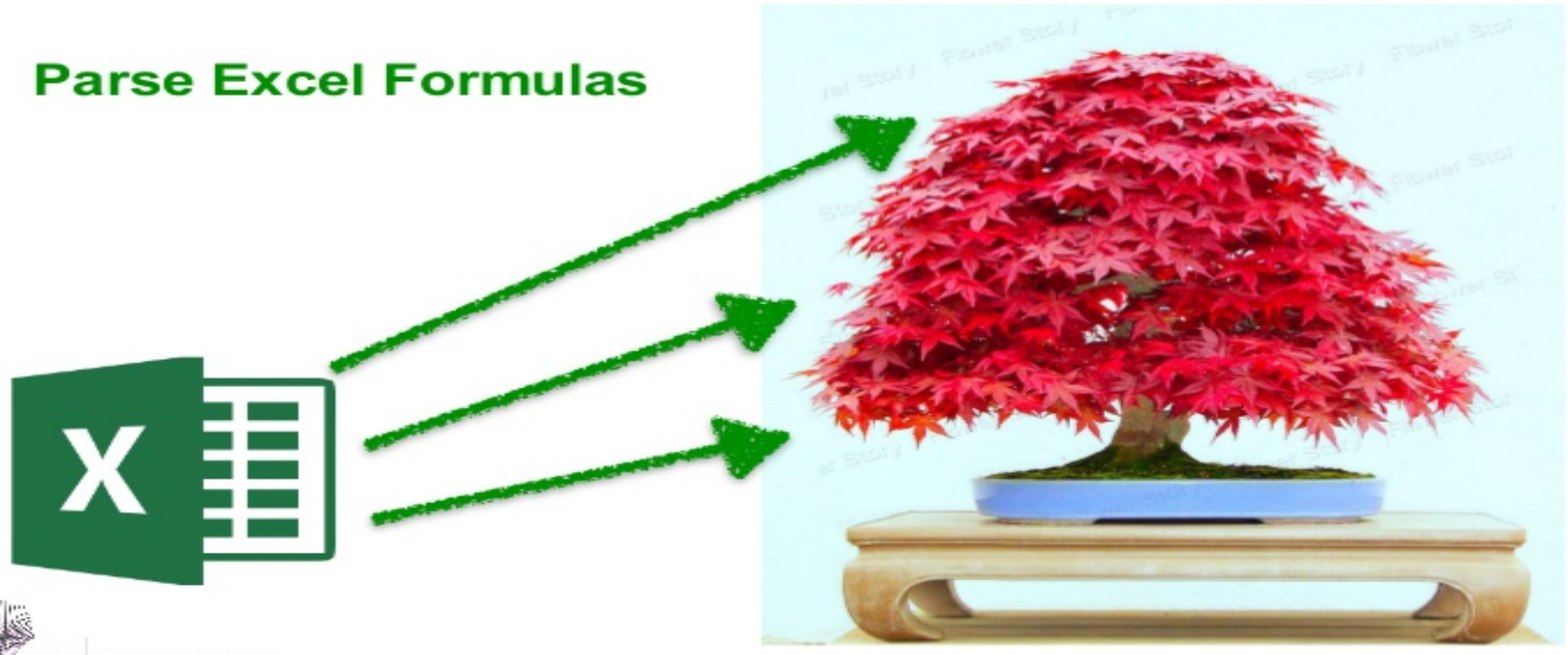
We need a Parse Tree!

Parse Excel Formulas



We need a Parse Tree!

Parse Excel Formulas

http://bit.ly/2eOTZAB

We need a Parse Tree!

Parse Excel Formulas

Generate Scala Code

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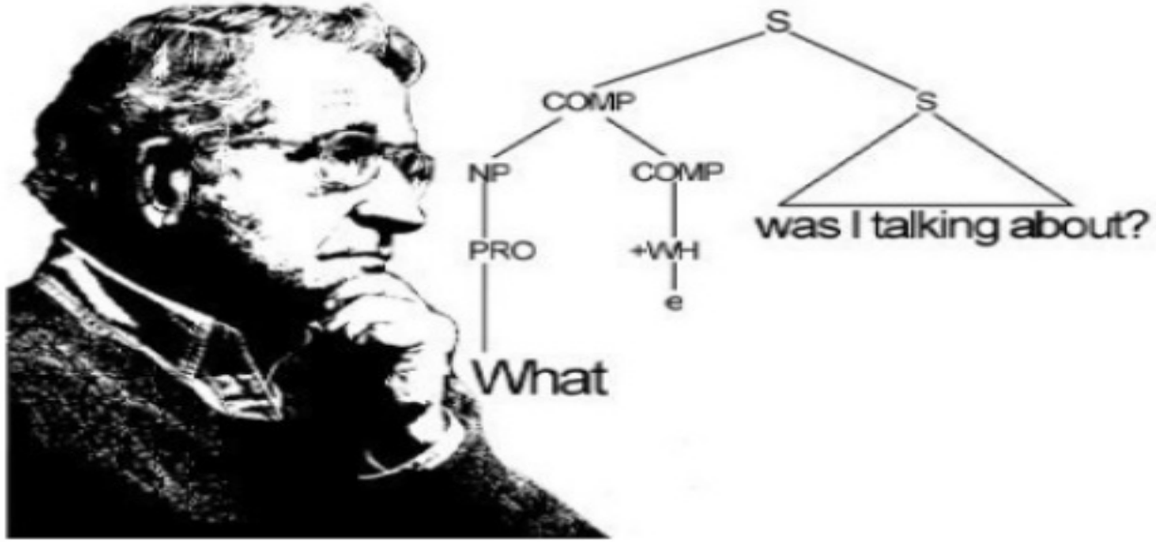
We need a Grammar!



http://bit.ly/2dH0ybF

XLParser

"If I have seen further, it is by standing upon the shoulders of giants"

— Sir Isaac Newton

A Grammar for Spreadsheet Formulas Evaluated on Two Large Datasets

Efthimia Aivaloglou, David Hoepelman, Felienne Hermans
Software Engineering Research Group
Delft University of Technology
Mekelweg 4, 2628 CD Delft, the Netherlands
e.aivaloglou@tudelft.nl, d.j.hoepelman@student.tudelft.nl, f.f.j.hermans@tudelft.nl

XLParser

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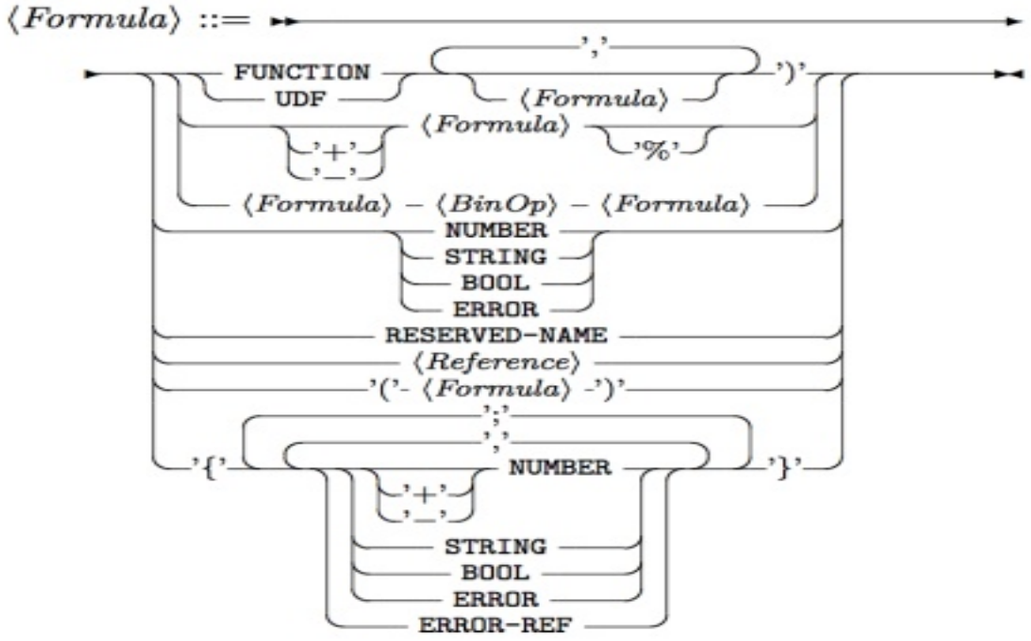

A Grammar for Spreadsheet Formulas Evaluated on Two Large Datasets – Efthimia Aivaloglou, David Hoepelman & Felienne Hermans, Proceedings of SCAM '15

XLParser

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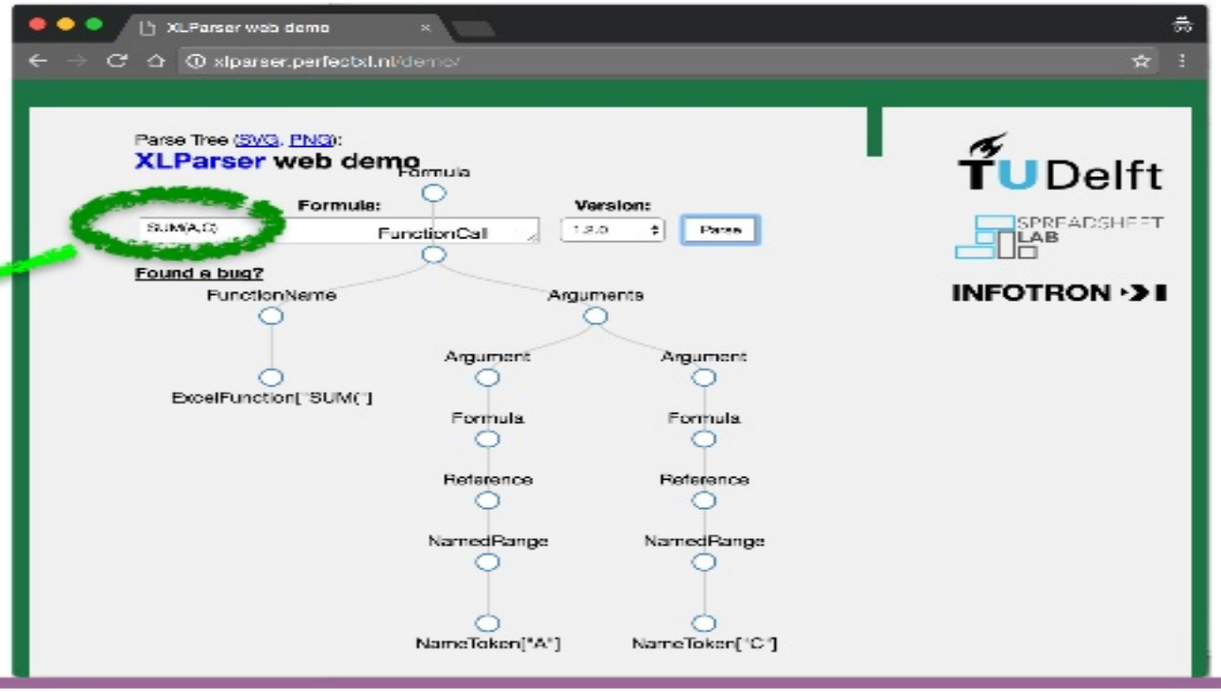
Excel Formula

SUM(A,C)



http://xlparser.perfectxl.nl/demo

XLParser

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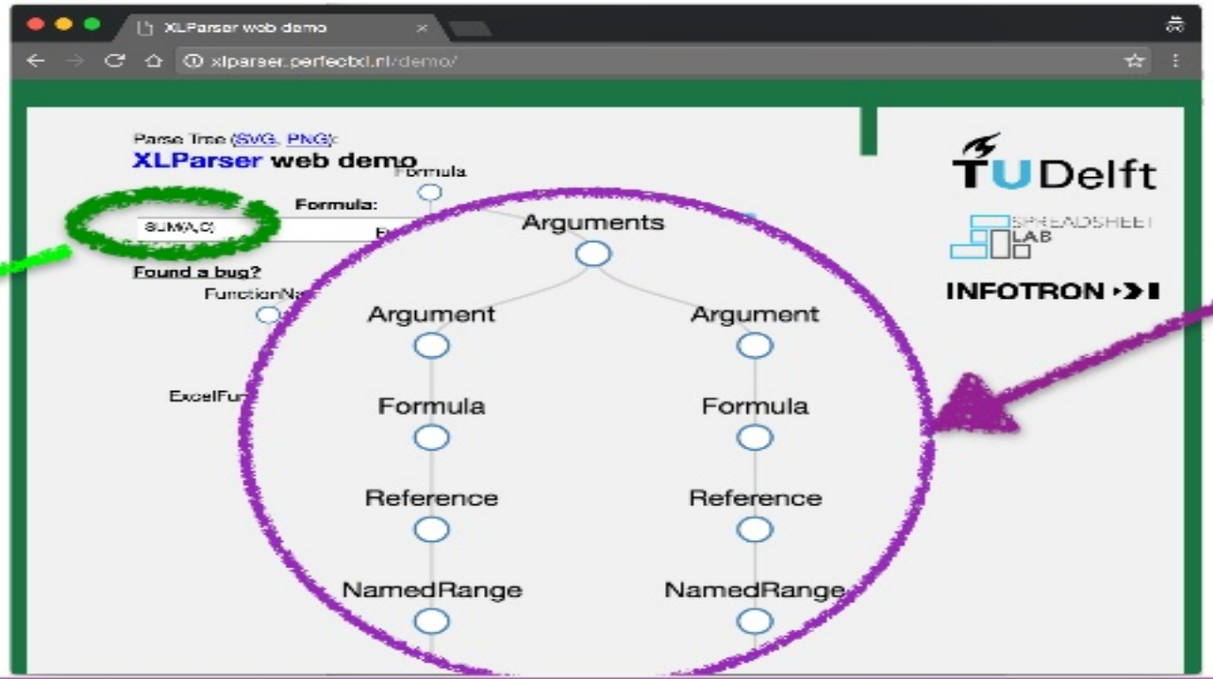
Excel Formula

SUM(A,C)

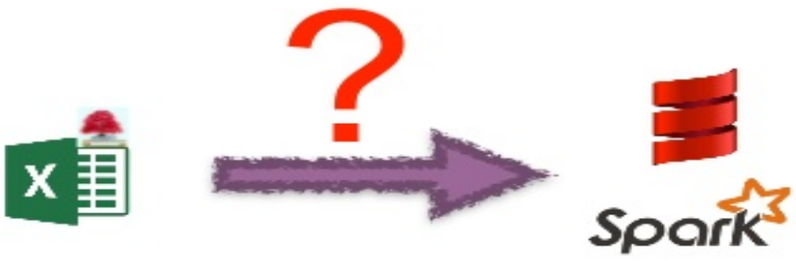


Parse Tree!

http://xlparser.perfectxl.nl/demo

Excel as a DSL

- External DSL: parsed independently.
- XLParser gives us a Parse Tree from an Excel Formula.
- Given the Parse Tree, generate code!



How do you generate code from parsed Excel Formulas?

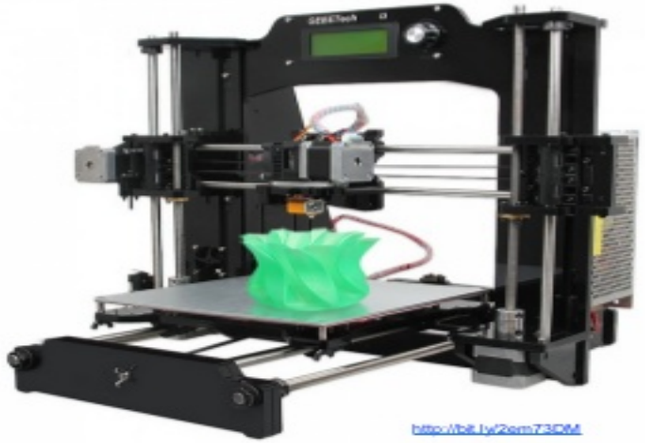
Generating Code

"An elegant way to generate code from an AST is to write a class for each non-terminal node in the tree, and then each node in the tree simply generates the piece of code that it is responsible for."

http://www.codeproject.com/Articles/26975/Writing-Your-First-Domain-Specific-Language-Part

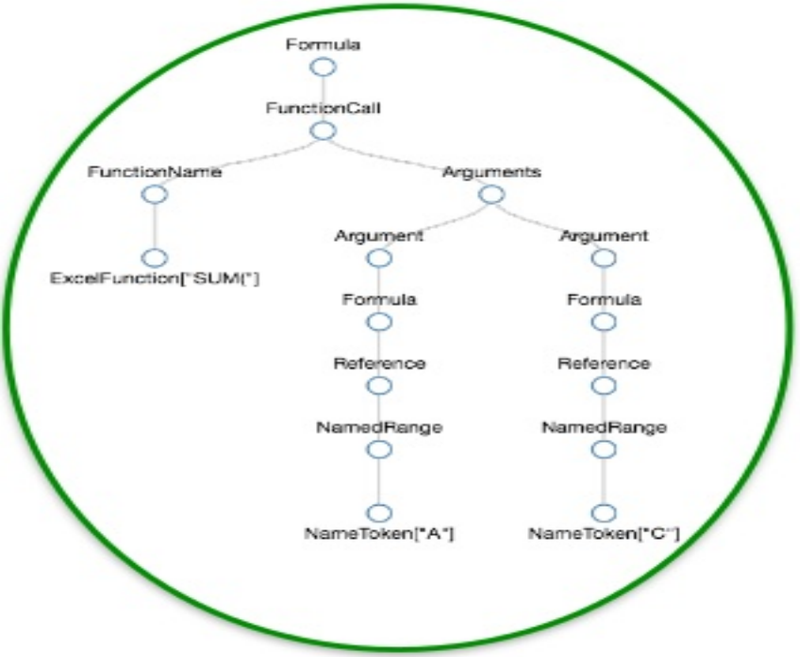
Generating Code

A practical way to generate code is to take a Parse Tree and write a pretty printer for the target language.



http://bit.ly/2ern73DM

Generating Code from an AST

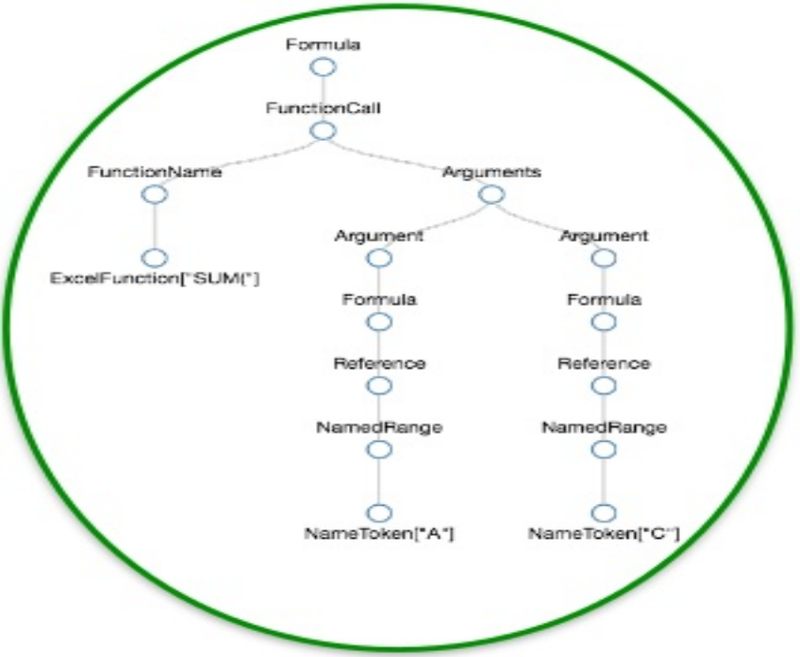


SUM(A,C)

```
> import org.apache.sql._
  import org.apache.spark.sql.functions._

  def SUM(A:Int, C:Int): Int = {
    return A + C
  }
  val applySUM = udf(SUM _)
  val sumDF = baseDF.withColumn("",
              applySUM(
                col("A"),
                col("C")
              ))
```

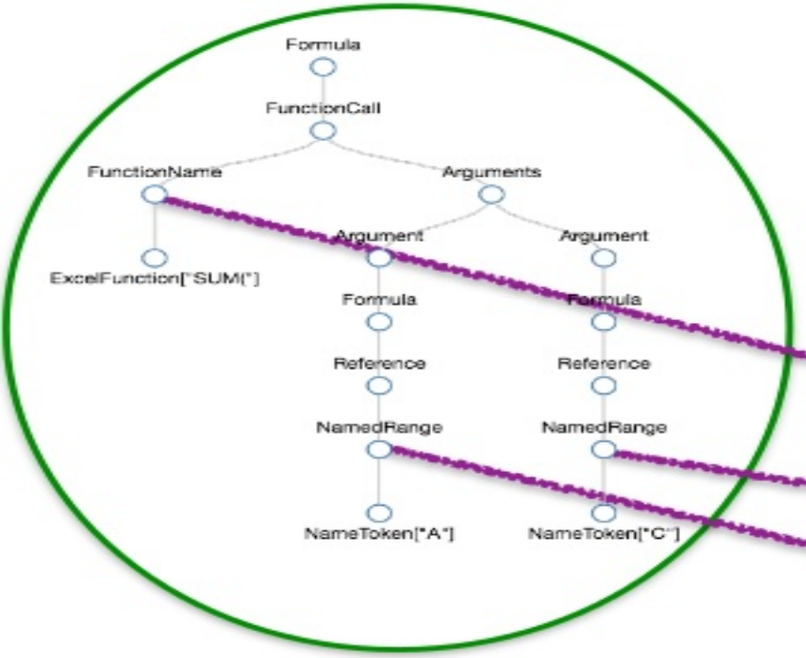
Generating Code from an AST




```
import org.apache.sql._
import org.apa[illegible]rk.sql.functions._

def SUM A:In[illegible] C:Int [illegible]Int = {
  return A +
}
val applySU[illegible]aseDF.w[illegible])
val sumDF[illegible]lumn("",
        applySUM(
          col("A"),
          col("C")
        ))
```

Generating Code from an AST




```
> import org.apache.sql._
  import org.apa  rk.sql.functions._

  def SUM A:In C:Int Int = {
    return A +
  }
  val applySU  aseDF.w  _)
  val sumDF  lumn("",
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          col("A"),
          col("C")
        ))
```
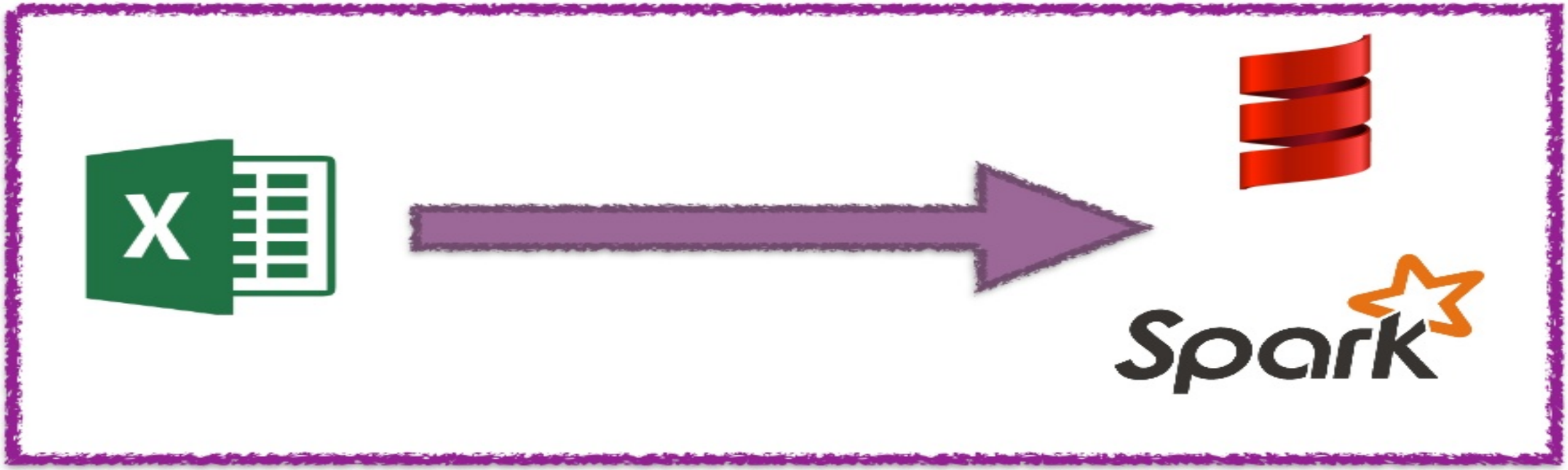

Demo!



What have we seen?

- Column-Oriented Excel Applications as Prototypes for Spark programs
- Program Transformation.
  - How to model as a Pipeline.
  - Why considered a Code-to-Code Transformation.
- How to Parse Excel Formulas.
  - Grammar
  - Parse Tree
  - XLParser
- Excel as a DSL.
  - How can we Generate Code?
- Demo.

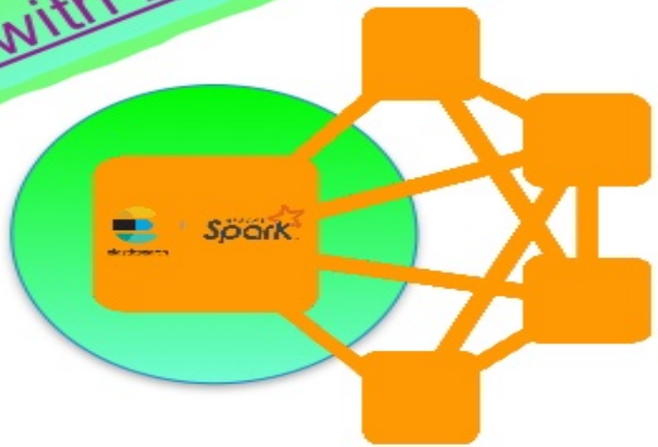
Next Steps

- Translate ~500 Excel Formulas.
- Modeling Machine Learning in Excel.
- Prototype D|'s and ML|'s in Excel.

Q&A

Q&A

Tomorrow: Spark Cluster with Elasticsearch Inside



THANK YOU.

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